

GIRONA COSTA BRAVA AP, Spain

WMO: 081840

Lat: 41.912N

Lon: 2.763E

Elev: 423

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	30.0	13.7	11.2	42.6	19.4	14.8	40.3	23.3	46.7	19.3	51.8	1.4	10	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	93.2	70.4	90.4	70.0	87.9	69.4	74.2	85.8	72.8	84.1	71.6	82.6	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	115.9	78.3	69.5	110.2	77.7	68.0	104.4	77.0	37.9	85.9	36.6	84.0	35.5	82.6	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.2	14.6	DB	22.2	98.1	2.8	3.3	20.2	100.4	18.5	102.4	16.9	104.2	14.9	106.6
			WB	20.9	76.8	3.1	1.7	18.7	78.0	16.9	79.0	15.2	80.0	12.9	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	46.2	47.2	51.7	55.9	62.5	70.6	75.3	75.4	68.8	62.7	52.7	47.4	(6)
	DBStd	11.64	5.00	5.13	4.96	4.67	4.98	5.35	4.29	4.22	4.66	5.31	5.91	5.15	(7)
	HDD50	440	138	103	39	7	0	0	0	0	0	2	39	112	(8)
	HDD65	2930	584	498	413	275	106	13	1	1	18	107	368	546	(9)
	CDD50	4006	19	24	92	182	389	617	785	788	563	394	122	31	(10)
	CDD65	1020	0	0	0	1	30	180	321	323	131	34	0	0	(11)
	CDH74	9880	0	0	12	46	377	1860	3254	3085	1006	238	2	0	(12)
	CDH80	3830	0	0	0	2	68	703	1424	1339	257	37	0	0	(13)
(14) Wind	WSAvg	4.5	3.6	4.0	4.7	5.4	5.1	5.2	5.4	5.1	4.4	4.1	3.7	3.4	(14)
(15) Precipitation	PrecAvg	27.2	1.9	1.7	2.2	2.4	2.4	1.7	1.2	1.7	3.3	4.0	2.5	2.2	(15)
	PrecMax	40.7	6.5	4.8	5.9	7.1	6.3	9.6	4.3	3.4	6.5	11.9	11.0	7.0	(16)
	PrecMin	16.3	0.0	0.0	0.1	0.1	0.2	0.0	0.0	0.2	0.6	0.5	0.0	0.0	(17)
	PrecStd	6.8	1.7	1.5	1.5	1.7	1.6	1.9	1.1	0.9	1.6	2.8	2.2	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	69.2	75.2	78.6	86.1	95.4	96.6	98.1	88.9	84.4	72.9	66.5	(19)
		MCWB	53.9	54.1	57.7	60.4	65.1	69.7	70.6	71.0	69.4	65.8	60.3	55.0	(20)
	2%	DB	62.6	64.6	70.3	74.8	81.4	90.7	93.0	93.2	85.7	79.5	69.3	62.9	(21)
		MCWB	52.6	52.8	56.0	59.4	63.7	68.9	70.6	70.8	68.9	66.2	59.0	53.9	(22)
	5%	DB	59.2	61.6	66.7	71.5	78.2	87.4	90.2	90.0	82.8	76.4	66.1	60.3	(23)
		MCWB	50.6	51.5	54.6	58.0	62.5	67.7	70.3	70.3	67.7	64.9	57.5	52.6	(24)
	10%	DB	56.2	58.6	63.7	68.1	75.1	84.0	87.8	87.4	80.2	73.1	63.2	57.4	(25)
		MCWB	49.3	50.0	53.1	56.3	61.4	66.8	69.5	70.1	66.9	63.5	56.0	51.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	56.4	60.1	62.9	68.0	73.6	75.9	76.5	73.4	70.5	63.6	57.5	(27)
		MCDB	63.5	65.1	70.9	74.9	80.2	87.6	88.4	88.0	82.6	77.5	68.8	62.6	(28)
	2%	WB	54.0	54.1	57.6	60.7	65.6	71.1	73.9	74.5	71.7	68.4	61.6	55.6	(29)
		MCDB	59.7	62.0	67.3	72.4	77.7	84.0	86.4	85.9	81.0	75.4	66.3	60.4	(30)
	5%	WB	52.4	52.5	55.8	58.9	64.0	69.5	72.5	73.2	70.2	66.7	59.5	53.9	(31)
		MCDB	57.9	59.7	64.0	69.2	75.3	82.9	84.8	84.0	79.1	73.2	63.9	59.0	(32)
	10%	WB	50.3	50.8	54.3	57.2	62.5	68.0	71.1	71.9	68.6	65.0	57.3	52.1	(33)
		MCDB	55.2	57.2	61.8	66.2	72.6	81.1	83.3	82.5	77.1	71.1	61.8	57.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	22.3	23.1	21.9	22.2	23.2	23.3	22.9	21.6	20.0	20.5	20.6	(35)
		MCDBR	25.3	27.0	28.5	27.5	27.0	28.0	26.7	27.0	25.2	24.1	23.1	23.0	(36)
		MCWBR	16.4	16.6	16.3	14.6	12.7	10.5	9.9	10.3	11.3	12.6	14.4	15.5	(37)
	5% WB	MCDBR	21.4	24.5	25.0	25.1	24.2	25.3	24.6	23.8	22.5	20.6	19.2	20.5	(38)
		MCWBR	14.8	15.8	15.1	14.4	12.7	11.0	10.5	10.9	11.5	11.9	13.0	14.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.314	0.332	0.377	0.402	0.421	0.431	0.437	0.432	0.410	0.383	0.342	0.314	(40)
	taud		2.522	2.471	2.357	2.296	2.290	2.294	2.296	2.316	2.363	2.430	2.495	2.525	(41)
	Ebn at Noon		266	278	275	275	272	268	265	263	261	254	252	255	(42)
	Edh at Noon		23	28	35	40	41	41	41	39	35	29	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		624	912	1309	1635	1913	2175	2155	1894	1445	1008	688	552	(44)
	RadStd		55	89	100	128	134	102	114	104	73	88	63	47	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+94

Nomenclature: See separate page